

Stroke physicians' and staff perspectives on machine learning to optimise thrombolysis decision making in stroke: a qualitative study

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Abstract

Background

SAMueL-2 (Stroke Audit for Machine Learning Project) working with the Sentinel Stroke National Audit Programme (SSNAP) developed clinical pathway and machine learning computer models to investigate variation in thrombolysis use. We investigated how this modelling could be designed and adapted to inform clinical practice and support optimal implementation of thrombolysis by exploring the perspectives of physicians and other staff whose work relates to acute stroke care.

Research Question

What should a machine-learning model based on SSNAP data look like, do, and deliver if it is to optimise improvement, and reduce unwarranted variation, in thrombolysis?

Objectives

1. To generate empirically and theoretically informed knowledge about how thrombolysis is currently delivered, centred on physicians' views, understandings, and practices.
2. To learn more about how stroke physicians' and staff think and feel about or use SSNAP, and about the use of machine learning in improving clinical practice.

Design and methods

We used focussed observations, semi-structured interviews and documentary analysis, to examine perceptions of thrombolysis, SSNAP, and machine learning. The Non-adoption, Abandonment, and Challenges to the Scale-Up, Spread, and Sustainability of Health and Care Technologies (NASSS) framework was used as a sensitising device to help us understand socio-technical factors likely to affect adoption and scale-up of SAMueL-2 technology.

Setting and participants

Hospitals were purposefully selected due to low rates of thrombolysis, differing stroke pathways, and for geographical variation. We conducted 184 hours of focussed observation in three NHS Trusts in England; comprising day/evening/night and weekend shifts and relevant meetings. We also observed online meetings of Integrated Stroke Delivery Networks (ISDNs) and other organisations with strategic overview of stroke services. 20 participants from the three observation sites and five key informants from other sites took part in semi-structured interviews.

Results

We present findings in relation to six NASSS domains: the condition, the technology, the value proposition, the intended adopters, the healthcare organisation, and the wider system. Our data showed participants were hopeful the SAMueL-2 technology could address variance in thrombolysis practice. It was seen as particularly suitable for junior clinicians, non-stroke specialists and at district general hospitals and offered value for training, reviewing clinical cases, and quality improvement.

Limitations

Our exploratory findings are not transferrable to all staff involved in acute stroke care/administration.

Conclusions

We identified three key learning points. First, given reservations expressed about SSNAP data, it is important to reassure intended adopters about the integrity of modelling based on this data. Second, evidence indicated ED physicians may have less confidence in the evidence base for thrombolysis. More work needs to be done with the ED physician community to build trust in the SAMueL-2 technology: recruiting ED physicians as brokers/clinical champions may address this. Third, perceived lack of funding and stroke workforce shortages may impede quality improvement and adoption of new technologies such as SAMueL-2. These concerns must be addressed to ensure sustained use and adoption. The next phase of the research will focus on the seventh NASSS domain relating to embedding and adaptation of the technology over time.

Background

Stroke is the fourth leading cause of death and the leading cause of disability in the UK ^{1(p164)}. Globally, it is the second-leading cause of death and the third-leading cause of death and disability combined ². There are about 100,000 strokes a year in the UK, and there are 1.3 million people living with stroke ³. There are marked health inequalities in the risk and incidence of stroke; stroke risk is linked to increasing deprivation and African/African Caribbean groups have double the incidence rate compared to the white population ⁴. Stroke survivors often experience residual and long-term consequences, both physical and psychological ⁵. Half of the people who experience stroke will have a resultant disability and a third will require assistance with daily living ⁶. Patel *et al* estimate that the number of stroke survivors aged 45 and over in the UK will increase by 123% between 2015 and 2035, with costs of stroke care expected to show a rapid rise over this period ⁵. The overall cost of stroke care in the UK is predicted to rise to £43 billion by 2025, and £75 billion by 2035 ⁴.

Eighty-five percent of strokes are ischaemic; caused by a blockage to the blood supply in the brain ⁷. Since 2002 thrombolysis, a medical treatment using a drug to break up a blood clot, has been approved in Europe to treat acute ischaemic stroke ⁸. Thrombolysis, provided it is delivered in the first few hours after stroke onset, has been shown to significantly reduce disability after a stroke ⁹ and is cost effective ⁴. The maxim 'time is brain' is used to emphasise that brain tissue is rapidly lost as the stroke progresses ^{10, 11} and the necessity of expediency regarding reperfusion (restoration of blood flow) treatment. Thrombolysis with alteplase is recommended for treatment of acute ischaemic stroke on the proviso that treatment should be started within 4.5 hours of stroke onset ¹². The 2023 National Clinical Guideline for Stroke for the UK and Ireland ¹³ delineated an extension of the time period in which thrombolysis can be administered: up to 9 hours in specific circumstances. However, thrombolysis carries some risk: it can induce a brain bleed in one in 25 people and this can be fatal in approximately one in 40 cases ¹⁴.

There is considerable variation in access to thrombolysis within and across European countries ⁸. An inverse care law is evident; where disability due to stroke is highest in middle age, thrombolysis use is often low ¹⁵. Currently in England, Wales and Northern Ireland about half of eligible patients receive thrombolysis though there is considerable inter-hospital variation ¹⁶. ¹⁷ The NHS Long Term Plan identifies stroke as a clinical priority and aims to increase thrombolysis rates so that all patients who could benefit from it (asserted to be about 20%)

receive it ¹. NHS England state that on average, for each patient receiving thrombolysis the NHS will save £47,000 over 5 years ¹⁸.

SAMueL-1 (Stroke Audit for Machine Learning Project) used novel machine learning approaches to clinical pathway simulation and computer modelling using data from the national stroke registry for the UK, the Sentinel Stroke National Audit Programme (SSNAP). This showed that about half of the inter-hospital variation in thrombolysis could be attributed to differences in patient demographics, and about half was the result of hospital processes and differences in clinical decision-making, the latter being the most significant factor ¹⁹⁻²¹. SAMueL-1 concluded that stroke thrombolysis rates of at least 18% were achievable in England and Wales, though local demographics would mean that individual hospital thrombolysis targets might vary between 5% and 26% ¹⁹. Qualitative semi-structured interviews with stroke physicians conducted for SAMueL-1 indicated that those from higher thrombolysing hospitals engaged more positively with the machine learning research, and those from low-thrombolysing sites tended to be more cautious ¹⁹. Findings suggested that stroke physicians had some doubts and concerns about the utility of machine learning in helping them to improve the stroke pathway, suggesting that further dialogue is required ¹⁹.

Subsequently, the SAMueL-2 project, working with SSNAP, continued to develop clinical pathway and machine learning computer models, to investigate variation in thrombolysis use in hospitals ²². The aim of SAMueL-2 was to investigate how a machine learning based approach could be designed and adapted to inform clinical practice and support the optimal implementation of thrombolysis. 'Explainable machine learning'²³ was used to understand which features affect the decision whether to give thrombolysis at different hospitals, and how each feature value affects the prediction. The machine learning has been expanded to predict outcome, and to understand what patient features most influence outcomes after stroke, with and without thrombolysis ¹⁶. The feature 'disability at discharge', recorded using the modified Rankin Scale, has been used to quantify patient outcomes after stroke. A Streamlit ²⁴ web application (app) with an interactive dashboard has been launched to allow hospitals to compare their decision making (which patients they would or would not treat with thrombolysis) with benchmark hospitals. The SAMueL-2 web app and the machine learning based analysis were developed and evolved alongside the qualitative component of the project described here. We refer to the machine learning based analysis and the app as the 'SAMueL-2 technology'.

To inform the design and implementation of the SAMueL-2 technology we explored the views, experiences and practices of physicians' and other staff whose work relates to acute stroke care. We used qualitative research methods to examine perceptions of, and attitudes to

thrombolysis, SSNAP and machine learning as separate but interrelated components. Additionally, we examined perceptions of stroke service delivery, organisational capacity to innovate, and the wider context of scope for quality improvement vis-à-vis thrombolysis.

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Theoretical Framework

We took a sociotechnical approach, seeing healthcare as comprised of networks of people, roles, organisational routines, technologies and systems, tools, documents, data and physical space ²⁵⁻²⁷. This encourages an empirical focus on the work practices and workflow into which a technology will be introduced ^{26, 28} and attention to how particular technologies are interpreted, perceived, used, or not used, and how this affects or is affected by the wider complex system ²⁹.

We used the Non-adoption, Abandonment, and Challenges to the Scale-Up, Spread, and Sustainability of Health and Care Technologies (NASSS) theoretical framework ³⁰⁻³⁴ as a sensitising device throughout data collection, coding and analysis. We chose NASSS because it allows identification and consideration of the multiple, overlapping influences (or 'domains') of influence on health-technology uptake and use. The NASSS framework is flexible and can be used prospectively to address implementation challenges ³⁴⁻³⁶ and has previously been applied to the implementation of AI/machine learning in healthcare ^{34, 36-39}.

Our analysis is also informed by the concept of boundary objects ⁴⁰ particularly in relation to the web app as the point of interface between the computer scientists and clinicians/stakeholders in acute stroke care. Boundary objects are epistemic artefacts that function to, 'inhabit several intersecting social worlds...and satisfy the informational requirements of each of them' ⁴⁰ (p 393).

Ayobi, Stawarz et al ⁴¹ have shown that machine learning explanations can function as boundary objects between computer scientists and users. Similarly, training materials have been used as a boundary object in the context of an assessment of a machine learning model ⁴². Boundary objects can facilitate communication ⁴³, develop coherence ⁴⁴ and act as mediators

and vectors of knowledge across different professional groups. Levina and Vaast ⁴⁵ suggested that ‘designated boundary objects’ become ‘boundary objects-in-use’ if they are usefully incorporated into local practice. The concept ‘emergent boundary object’ ⁴⁶ has been used to reflect the processual nature of designing artefacts and is informative in the context of the ongoing development of the SAMueL-2 technology .

Methods

We used a combination of observations, interviews and documentary analysis to gain insight into stroke care and the potential use of the SAMueL-2 technology ^{25, 26, 47}.

Observation of healthcare professionals and work practices involved in the acute stroke pathway in three NHS Trusts (acute hospitals) in England (hereafter referred to as Sites A, B and C) was conducted by researcher KPB. We used SSNAP ⁴⁸ data to purposefully select hospitals with low rates of thrombolysis and ensured they had differing stroke pathways and were geographically dispersed. KPB carried out 184 hours of focussed observation across the three sites comprising a variety of day/evening/night and weekend shifts, observing stroke care as well as clinical governance and multi-disciplinary team stroke review and SSNAP review meetings. Additionally, observation of online meetings of Integrated Stroke Delivery Networks (ISDNs) and other organisations with strategic overview of stroke services in the NHS took place, with a focus on thrombolysis. The observations were informed by the NASSS framework ^{31, 34} which was used to sensitise ⁴⁹ (insert file). Contemporaneous fieldnotes were made.

Additionally, KPB carried out semi-structured interviews with 20 participants from the three observation sites and five key informants⁵⁰ (senior clinicians/managers involved in stroke initiatives) based at other sites (see table). All interviews were one-to-one except one interview with two participants. Interviews were conducted face to face or via video call, depending on the participants’ preference. Interviews explored the views of clinicians and other professional staff involved in acute stroke management/administration. They focused on experiences of, and perspectives on, thrombolysis and views on improving practice vis-à-vis thrombolysis, stroke national audit data and machine learning. Interview topic guides, developed by the project team and informed by the NASSS framework were used (insert file). Verbal consent was obtained for observations, and written consent was obtained for interviews. Interviews were audio-recorded and transcribed by a GDPR compliant transcriber and checked for accuracy by RJ. We have not specified participants by site to avoid potential identification.

Interview Participants	
Consultant	11 Stroke, 3 Emergency Department (ED), 1 Care of the Elderly
Associate Specialist Doctor (ED)	1 ED
Registrar	1 Stroke, 1 ED
Stroke Assessor & Nurse	2
Stroke Nurse	1
Advanced Clinical Practitioner (Stroke)	1
Stroke Unit Administrator	1
IT Co-ordinator (Stroke)	1
ISDN manager	1

A third source of data was documents: we collected Trust-level documents (such as thrombolysis protocols and quality improvement (QI) initiative documentation); policy and strategy documents at ISDN level; computer modellers' presentations to stakeholders (clinicians, senior managers, policymakers); modelling team summaries of feedback from stakeholders; SAMuel-2 research team meeting minutes; and SAMuel- 2 Github pages ⁵¹.

Sampling and Recruitment

We used a purposive sampling strategy ^{52, 53} to recruit NHS staff ensuring a range of professional roles, grades and experience and knowledge of thrombolysis. Purposive sampling of professional networks enabled recruitment of physicians and staff from sites outside the three selected for clinical observation, to maximise applicability and variation of the sample. The sampling strategy was guided by the notion of information power, with the premise that the more apposite information the sample holds, the fewer participants may be required ⁵⁴. Potential participants were initially identified by SAMuel-2 team members' existing contacts at the participating sites and by the Clinical Research Network (CRN) across a diversity of NHS trusts. Further recruitment of staff involved in thrombolysis took place during fieldwork.

Patient and Carer Involvement (PCI)

The SAMuel-2 project is supported by a dedicated Patient and Carer Involvement (PCI) group. In relation to this aspect of the project they contributed to the qualitative research design, reviewed project materials, and provided feedback on findings from patient and carer perspectives. These contributions took the form of discussion in meetings in which researchers presented materials to the PCI group for discussion and through the membership of a PCI group representative on the study steering group.

Ethical Approval

HRA & HCRW approval was provided by the Essex Research Ethics Committee in August 2023 (IRAS 322303; REC reference 23/EE/0124).

Data Analysis

We imported and managed data in NVivo 14⁵⁵. Data-analysis followed an abductive logic⁵⁶ which involved moving back and forth between examinations of the data, theories, and concepts to develop our interpretation whilst seeking to identify and code exceptions and contradictions⁵⁷ that helped us to see where our interpretation needed more work. We used the NASSS framework as a sensitising device that informed the analysis but did not determine the scope of the findings⁵⁸. The NASSS framework has been thoroughly and extensively operationalised³⁰⁻³⁴ and the key concepts/issues comprising the various domains informed and sensitised the deductive qualitative analysis^{57, 59}. Coding involved a process of moving between careful reading and re-reading of the data, coding deductively with reference to the NASSS domains and inductively by intentionally searching for alternative ideas or concepts relevant to the research questions that were not predicted within the framework^{60, 61}, and which could be categorised into a code. Coding was initially carried out by RJ, with discussions of the coding and thematic analysis held with IL and CP to develop the analysis presented here.

Results

We present our findings here in relation to six NASSS domains: the condition, the technology, the value proposition, the intended adopters, the healthcare organisation, and the wider system. The seventh NASSS domain relates to embedding and adaptation over time, focussing on scope for adaptation of the technology over time and organisational resilience and this will be a focus of the next phase of the research.

Domain 1 The Condition

Physicians emphasised the heterogeneous and complex nature of the presentation of patients with stroke. They frequently used the concept 'barn door' to refer to a situation when a patient was perceived to be presenting with 'classic' symptoms of ischaemic stroke and where guidelines for treatment could easily be followed and decision-making was straight-forward. However, physicians suggested most patients were not 'barn-door' presentations, making decisions regarding thrombolysis more challenging:

I mean, if it's a barn door situation, most people would thrombolyse. So, I don't think that is the issue we have.

Stroke Consultant 4

... the straightforward [thrombo]lysis decisions are... are far apart, you know? The majority of patients that come to hospital are much more complicated than the person who just comes in and is like, "Oh, I've had my..." you know? "I had my stroke half an hour ago, and all I've got is weakness down one side. I'm not on any medication. You know? I fulfil all the criteria."

Stroke Consultant 6

Clinicians stressed the complexity of decision-making with respect to thrombolysis, particularly in clinical grey areas, and the perceived level of risk involved in the procedure. They discussed weighing up the risks/benefits in each case and challenges of obtaining informed consent for thrombolysis:

So, if you've got a patient, for example...we've got a NIHSS [National Institutes of Health Stroke Scale] of four, five. Can just about lift the... can lift the right arm and leg, speech is normal. So, you could thrombolysed them. But you can also thrombolysed them and cause a catastrophic bleed and you've got somebody who's talking to you, and just can't move their right arm and leg very well. Got a catastrophic bleed and they've died. So, thrombolysis, you know, is not straightforward in terms of, there's a risk of bleeding. And thing is, is do no harm. You know?

Stroke Consultant 5

One said that decision-making was becoming more complex with the introduction of new clinical guidelines for stroke ¹³:

...it's suddenly got a lot more complicated. So, there's calculating the midpoint to sleep, there's doing lots of additional imaging, there's looking at CT perfusion, or MRI scans and seeing whether there's a mismatch. And so, there is a whole load of extra judgement calls that you have to make on individual patients which, before when it was everyone before four and half hours, yes, everyone after four and half hours, no. And that's pretty much it...But it's gotten more complicated than that. So, I think that's what makes it tricky.

Stroke Consultant 3

Clinicians characterised themselves and colleagues as pro or anti-thrombolysis, or keen or reluctant thrombolysers, the latter typically because they had experience of negative patient

outcomes following thrombolysis and/or were generally more risk averse. Some reservations were also expressed about targets to improve thrombolysis rates.

It's quite a difficult intervention to... to believe in when your last two patients have died from the... the intervention you've done.

ED Consultant 1

It's about experience and it's about individual consultant style, how risk averse they are. How many patients they've seen bleed. What their general feeling is about people's... what are we trying to achieve here? Are we trying to get thrombolysis rates...? Are we trying to give something just for the sake of doing it? Or are we looking holistically at the patient?

Stroke Consultant 5

Domain 2 The Technology

SAMueL-2 involves machine learning models based on SSNAP. Participants had differing understandings and experiences of SSNAP and of machine learning-based modelling, and of how they might come together.

Participants from one of the sites, emphasised the necessity of the SAMueL-2 technology incorporating data on the configuration of acute stroke services for it to be seen as having validity:

This Streamlit app, you need to build in the key organisational audit statistics from all the hospitals to say this service is led by this bunch of clinicians, whereas, this service is led by this bunch of clinicians. And they're totally different. One's a specialist discipline, one's a generic A&E consultant discipline...

IT Co-ordinator 1

Most participants expressed having trust and confidence in SSNAP data and some referred to its potential to improve stroke services, but not all understood or engaged with the programme and thus were unsure how machine learning from SSNAP data could improve thrombolysis decision-making:

It is a... it is a huge set of data, isn't it?...So yes it can be very useful. It is such a great collection of data. You have almost everything there, from the patient characteristics, ethnicity, to their comorbidities and risk factors and anti-coagulations and everything...

But how it would help you in making decisions regarding thrombolysis in the future, it's difficult for me to understand, honestly speaking...

Stroke Consultant 8

Others, were more enthusiastic about the potential for machine learning to extend the utility of SSNAP as a learning system for quality improvement:

...because SSNAP is a good quality reporting system, but it's not a learning system. Well, it is kind of a learning system, but only when it's used by the units to say 'this is the area. What are we learning from that?' And here, you might have different ways of... of transferring it more into the learning direction.

Stroke Consultant 7

The addition of machine learning to the SSNAP portal was mooted as improving its sensitivity and potential for quality improvement vis-à-vis thrombolysis.

I think from a user interface that would be brilliant. Basically, a chat GPT using the SSNAP dataset.

ISDN Manager 1

A few interviewees discussed concerns about SSNAP data quality and questioned the assurance processes for these audit data.

Because it's human-generated data, it can be open... I don't think we're to blame for this...I don't think we do this...[but] when you have an... you know, a non-automated system, with human input, it then opens itself to abuses.

Advanced Clinical Practitioner Stroke 1

Some clinicians discussed their current use of AI based systems such as Brainomix⁶² or RapidAI⁶³ and showed interest in SAMueL-2 becoming interoperable with these systems. Others expressed scepticism:

So, what I really love is to have a tool that talks to, say, the current electronic system that we use for scans. So we call it Brainomix, and whether it can be used in the acute phase to make a decision or whether you then use it later on when you're reviewing the case, if that, whatever, if SAMueL could talk to Brainomix or... and whatever system is used to get the scans and then amalgamate it with the clinical information that we've got, that would be amazing.

Stroke Consultant 1

Well, we've got RapidAI, which is the machine learning for needing a CT scan, CTAs, and CTP. I must admit we're a bit sceptical about the output sometimes. So, we're not completely 100% trustworthy of it.

Stroke Consultant 6

Domain 3 The Value Proposition

Reporting in this NASSS domain focusses on perceived demand-side value (while perceived supply-side value is reported as part of Domain 6 'The Wider System').

As defined by Alami, Lehou et al ³⁸ in relation to AI technologies, SAMueL-2 can be characterised as being at the *value promise* developmental stage; our data reflects the perceived or anticipated value of the technology which is relatively speculative ⁶⁴.

SAMueL-2 was seen by some as providing the potential to address variation in clinical practice between clinicians and across trusts:

It should be a good way to... level things out, shouldn't it? And have... us all making more similar and clinically correct decisions, because we shouldn't have the same decisions based on the... same clinical decision based on the same guidelines nationwide, yet we know that some hospitals are thrombolysing 20 to 30 percent of people, and some hospitals are thrombolysing two percent of people, [it] doesn't make sense. So, [SAMueL-2] sounds like a very good way in terms of trying to work out why that is.

Care of the Elderly Consultant 1

Many participants perceived value in the potential for machine learning to help them in clinical grey areas when decision-making was most difficult, but there was uncertainty and ambivalence regarding whether this would/could happen during the acute stroke pathway. They hoped the SAMueL-2 technology would provide more specific and objective assessments of risk/benefit of the procedure and inform consent discussions.

We should have the tool for grey areas where there is no clear guidelines, an NIHSS score of less than four, or probably last seen well at five o'clock, and now presenting at ten o'clock, so ten hour... five-hour's difference, which is just beyond.

Stroke Consultant 8

Some perceived it to be inevitable that AI/machine learning would eventually be used in real-time at the point of care:

Eventually, with anything that's like this, is that it will come out, you know, as a front-door tool, to assist... That's not particularly the premise now, is it? But that will... I think that would be the next step from here..to assist with the decision to thrombolysed.

Advanced Clinical Practitioner Stroke 1

Some clinicians saw the value of the benchmarking feature of the web app to change culture or local thrombolysis guidelines. However, it was emphasised that this benchmarking should include data on patient complications and outcomes for it to be seen as valid and useful.

It would help particularly with those when you're trying to shift the beliefs of physicians, because if you feel that this patient's really not going to benefit, "I'm not going to do it." Then you know... I tell you that there is a machine learning and an AI to tell you, "Look, actually, you're wrong, the vast majority of stroke physicians would actually thrombolysed."

Stroke Consultant 4

So, it might speed up the not so early uptakers and saying, 'here's the data. 70% do it... but you would need to have data saying, 'and that's their complication rates" or saying, 'this is a safe thing to do'. Because if you see independent data and people do this, then it might not take you five years to convince people here that we need to change the guideline.

Stroke Consultant 7

Participants discussed the perceived utility of the SAMueL-2 technology as a 'learning tool' to review clinical cases and provide training or quality assurance, for example at governance meetings. It was perceived as being useful as a decision-aid for trainees, non-stroke specialists and at district general hospitals (DGH).

I really want the SSNAP to inform the AI and the machine learning to say, 'Look, this is what's happening'... So, have a list of patients... so, patients who have not been thrombolysed, why have they been not thrombolysed? Could they have been thrombolysed? And pick up sort of very similar sort of patients across the country to say, let's just pick up this particular type of patient in 10 centres, or 100 centres, and 10 physicians didn't thrombolysed, but 90 did. So, you have sort of case scenarios built into the machine tool, so, you know, the machine tool sort of then provides that example to

say, “Look, this is what’s been done. What has happened. And this has been gathered from this new data... SSNAP data.” Then, we could confidently sort of work... deliver that if we were not a firm believer of it initially.

Stroke Consultant 4

Domain 4 The Intended Adopters

Some Emergency Department physicians were less confident about the evidence base for thrombolysis. They expressed a lack of faith in the clinical trials and reluctance to thrombolysise due to perceived risks of the procedure. One said they preferred thrombectomy to thrombolysis because they perceived that thrombectomy had more robust evidence to support it and fewer risks associated with it:

I think it's a hard decision because there are quite a lot of risks as well as the possible benefits. I feel much happier when people can go for thrombectomy. I think it's an easier decision when they're outside the thrombolysis window, they can go for thrombectomy. I feel like we're still doing something for them...I think thrombectomy to me seems to make more sense with the evidence base without the associated risks of thrombolysis.

ED Consultant 2

Some interviewees asserted that the benchmarking feature of the web app was not suitable or useful for experienced consultants like themselves, emphasising trust in their own clinical acumen:

I'll be very honest with you, I wouldn't be too bothered... bothered with what the other centre is doing. I'd want to do what I felt based on the evidence in front of me, and... and no disrespect to the other centres. Because it may be some of the juniors, you know, like the DGHs [district general hospitals] may not be thrombolysing that person who I might actually thrombolysise.

Stroke Consultant 10

Others worried about how benchmarking and comparison might be used and were anxious about how it might affect them:

“Would the thrombolysis tool be used to hold over you, instead of being there to help you? Could it be a double-edged sword?”

Conversation with stroke clinicians, Fieldnotes 11/11/23.

Concern was also expressed that AI might be given primacy over clinical decision-making and that this could adversely affect patient outcome:

I know that's what AI's meant to replace...the clinical Gestalt which is ...your clinical Gestalt feeling, and what you've picked up, and...your feeling about a case. But again,...AI is kind of quite a difficult one when it's kind of asking you to do something, or advising you to do something, if you then do it and it goes wrong...and your own Gestalt was saying, "Hmm, should I? Should I not?" That's quite a difficult one to... morally speaking to kind of reconcile. Because you know, if a machine... says, yes, push the plunger, and I do it, and the patient bleeds... and dies, machine's not going to think...about that. I'm the one that's going to... bed that night...knowing that I've just killed somebody... I think [we] ignore human beings at our peril.

ED Consultant 1

Domain 5 The healthcare organisation

Participants discussed recent successful quality improvement initiatives regarding stroke, for example, improvement in door-to-needle times. For example, site A had initiated a stroke quality improvement week, and Site B had undertaken a reconfiguration of the acute stroke pathway which involved new stroke nurse assessor roles and stroke healthcare assistants. Each team member had been designated specific tasks facilitating speed of the acute pathway. Most participants at site B felt confident in the acute stroke processes and that there was readiness for organisational change. One participant suggested that there was potential for experienced non-consultant level staff to thrombolysed without a consultant being present to speed up the process.

However, participants also described organisational-level barriers they perceived to be impeding thrombolysis provision. Various organisational issues were identified as problematic:

Inexperienced clinical leadership, or people who have been clinical... either been clinical leaders for too long, or too short. General culture in the organisation of risk aversion. Bureaucracy of having to report things. Terror of going to Coroner's Court and having to stand up and justify yourself. You know? And then a reeling off of kind of clinical reasons why... these patients should, these patients shouldn't...I think probably the single biggest factor is that we have got teams that are led by people and have got a culture of doing it, and then teams that don't.

ISDN Manager 1

Some people were enthusiastic that SAMueL-2 technology could be instrumental in overcoming institutional barriers:

I think anything which can help us break barriers with information which is standardised information, and which is helpful to show what good practice should be and how it can be done in other areas, that is something for... for barrier breakdown. Because the barriers within NHS organisations are still there. They are very much there.

Stroke Consultant 7

At site B issues cited as barriers to improving thrombolysis included the limited availability of CT perfusion (8 –11 am) and delays reporting scans overnight due to outsourcing of this service. Similarly, lack of access to imaging was a barrier at Site A, particularly with respect to the extended time window for thrombolysis in new guidelines¹³. Participants mentioned lack of trained personnel available to interpret scans and delays waiting for reports.

We don't have the advanced imaging here to look for perfusion scans and MRIs to look for a number that could be saved, etc. So, at the moment, we're very much limited to just the broad-brush strokes of what are the... the overall statistics of harm versus benefit.

ED Consultant 3

In both Site A and C clinicians discussed stroke workforce shortages such as a lack of specialist nurses, particularly at evenings and weekends. At Site C difficulties in recruiting registrars and high staff turnover were also cited as problematic. At Site A the shortage of stroke specialist physicians was considered to be constraining thrombolysis decision-making and delivery:

I think here, because the clinicians making the decisions aren't stroke specialists and actually, sometimes they're not doing it that much because there's loads of ED consultants so they're not seeing that many strokes each... Often it's like, 'they've got an exclusion or a relative contraindication, so we won't do it.' So, it's a lot more... I think they stringently adhere to the guidelines here including relative contraindications, usually meaning that people aren't thrombolysed. Whereas perhaps if you had a more experienced... whether it's a stroke physician or an ED physician with a, you know, particular interest in stroke, if you've got someone that's a bit more confident with a bit more experience, then we would probably thrombolysed more people here.

Care of the Elderly Consultant 1

Clinicians in Site A stated that there were stroke improvement plans in place, but that workforce, infrastructure and funding constraints presented significant challenges and they perceived change to be difficult. Similarly, clinicians at Site C described over-crowding and lack of sufficient access to computers in acute stroke treatment in the Emergency Department and perceived organisational change to be a slow process.

Says A&E is “not fit for purpose, you’re fighting for phones, computers, beds ...4 of us standing up around the computer.” ... says it would be impossible to open up a tool to look at in this kind of environment.

Conversation with stroke nurse, Fieldnotes 21/11/23

Domain 6 The Wider System

Our analysis indicated that the wider professional and political context was important for adoption and scale-up of SAMueL-2. Stakeholders and participants from ISDNs were, with some expressed reservations about usability of the SSNAP interface, enthusiastic about SAMueL-2 being added to the portal and about facilitating its roll out to and uptake by sites for quality improvement.

Run it through the SSNAP website really. I mean, the SSNAP website is not the easiest to use to some degrees. I’ve tried... struggled to find a few things on it. But yeah, you could just have a portal on the on that...I mean, the ideal way of delivering it would be probably through the ISDNs, and it’s certainly initially around QI, and around rolling it out...and there’ll be sites that are all over it, and they love that sort of thing. And then, I think probably where the networks add more value is where those...those that are possibly underperforming and just not engaged, about how you chip away at that...

ISDN Manager 1

Future roll out of the SAMueL-2 technology was discussed in the context of, “some money to do some QI work through NHS England” and work by the research team with the ‘Thrombolysis in Acute Stroke Collaboration’ (TASC), to improve access to thrombolysis. At the time of writing, six hospitals were involved in beta testing of the SAMueL-2 web app as part of this initiative.

The development of machine learning in conjunction with SSNAP was referred to by one national level policymaker as offering, “the prospect of contributing significantly to reductions in stroke-related disability” [stakeholder communication] and was seen as having relevance to the NHS Long Term Plan. This appears to constitute a clear supply-side value proposition. The

TASC initiative provided a ‘policy push’ for SAMueL-2 technology implementation, albeit initially on a limited scale.

Discussion

Our aim in this study was to investigate current thrombolysis delivery, use of national audit data on stroke care, and the potential for machine-learning to improve practice. Many participants perceived value in machine learning based on SSNAP, hopeful that it could address variance in thrombolysis practice. However, possibly as the SAMueL-2 technology was under development/evolving it appeared to be an ‘emergent boundary object’⁴⁶. Participants were unsure if/how it might work in practice/clinical workflows. There was ambivalence with respect to its potential use as a real-time decision-aid; regarded as less acceptable to experienced clinicians. This is concordant with previous research showing concerns regarding professional autonomy affected the acceptability of computerised decision support systems/AI to clinicians^{37, 38, 65}. The SAMueL-2 technology was seen as suitable for junior/inexperienced clinicians/non-stroke specialists/at DGHs and offered value for training, reviewing clinical cases and for quality improvement. The benchmarking feature of the web app was discussed as potentially culture changing, operating as a boundary object facilitating consensus regarding thrombolysis practice. At the time of writing the web app is undergoing further development and testing and questions regarding inclusion of clinical outcome data and interoperability are being explored. This will likely prove important as lack of interoperability of AI technologies with an organisation’s systems have been shown to be a significant obstacle to their use³⁸.

Our analysis indicated perceived supply-side value of the technology and efficacy as a boundary object ‘speaking’ to policymakers and engendering ‘techno-optimism’³⁸ with respect to its capacity to reduce stroke related disability. Three specific challenges to its use in practice and wider scale-up and sustainability of the technology are highlighted by our analysis.

First, some intended adopters expressed concerns about the quality of SSNAP data. These concerns are echoed in the Stroke Getting It Right First Time Programme National Specialty Report (GIRFT)^{66 (p 12)}. Our exploration of views about the SAMueL-2 technology adds to a corpus of research highlighting reservations about data quality, or broader lack of awareness of data collection and reporting and the variable social practices (and inaccuracies) in data reporting⁶⁷⁻⁷¹. GIRFT^{66 (p 24)} recommendations to strengthen audit and review by formalising ISDN assurance processes for quality of SSNAP data entry may allay concerns regarding SSNAP data integrity and thus support the roll out of the SAMueL-2 technology. It will also be important to show intended adopters that modelling is robust to any possible errors in SSNAP

data, by showing that we are looking for general patterns that are consistent across the data set.

Second, our analysis indicated that some clinicians, notably ED physicians, were more wary of initiatives to increase thrombolysis. According to the 2016 SSNAP Acute Organisational Audit ^{72p40}, 10 percent of sites had an ED physician on the on-call thrombolysis rota. Given the shortfall of consultant stroke specialist provision in the UK ⁷³ non-stroke specialists are likely to continue to be involved in thrombolysis decision-making and delivery. Evidence shows the wider ED physician community of practice has some reservations about the efficacy and safety of thrombolysis for acute stroke ⁷⁴⁻⁸⁴. Dewar *et al*, noted ^{75 (p 8)}, ‘Emergency physicians appear to feel uncomfortable diagnosing stroke and so fret about causing unnecessary hemorrhage in patients who should never have received tPA in the first place’. Similarly, Langlois-Thérien *et al* ⁷⁹ argue that emergency physicians’ different ‘epistemic evaluation’ process means they do not accept the evidence base for thrombolysis in stroke in the way the neurology community of practice does. The Royal College of Emergency Medicine position statement from 2015 about acute ischaemic stroke and intravenous thrombolysis ⁸⁵ stated:

The College supports robust governance procedures, but does not support the translation of elements of these into arbitrary performance targets or commissioning targets (for example, regarding the number of patients actually thrombolysed). This is a concern that this may create perverse incentives within stroke care, or be prey to the Goodhart principle.

Given this background and reservations expressed by ED physicians in our study this requires further investigation. We are unsure whether the SAMuel-2 technology could become a ‘boundary object-in-use’ ⁴⁵. It seems likely that there may be some resistance to use of the SAMuel-2 technology and it may be that brokers ^{43, 86}, or ED physicians as champions may be required to support adoption ^{87, 88}.

Third, our analysis showed there are organisational challenges to the integration of SAMuel-2 technology and improving of thrombolysis rates. Participants informed us that lack of funding and workforce shortages impeded care and organisational change. Other sources document shortages of staff working in stroke care ^{11, 66, 89-91} and how a lack of specialist staff is asserted to be compromising the ability of the NHS to deliver the latest medical advances and best treatment to stroke patients ⁸⁹. A recent evaluation of an e-stroke technology (Brainomix) ⁹² showed staff perceived organisational culture and staffing composition to be factors that could hamper or accelerate effective adoption of the technology. Participants in our study also cited

lack of availability of requisite imaging, for example, CT perfusion, as constraining thrombolysis, and said this should be addressed as a priority. Variability of access to appropriate imaging across sites treating acute stroke is a common concern ^{66, 92, 93}. Lack of these kinds of key resources (services and/or staff) could hinder scale-up of SAMueL-2 technology.

Our research was qualitative and exploratory and therefore findings cannot be said to be representative of the views of all staff involved in acute stroke care/administration. Nonetheless, our theoretically informed research has provided some key insights relating to potential challenges to adoption and scale-up of SAMueL-2 that are corroborated by extant research. Testing of the web app currently underway by hospitals involved in TASC will provide further information relevant to the value proposition/implementation.

Conclusions

In this study we drew on the NASSS framework to help understand the socio-technical factors likely to affect the adoption and scale-up of SAMueL-2 technology. This research adds to a relatively small but growing body of knowledge from a socio-technical perspective and using qualitative methods to investigate challenges to the implementation of artificial intelligence/machine learning in healthcare ^{34, 36-39}.

We identified three learning points which may facilitate further implementation of the technology. First, given reservations expressed by some of our participants and healthcare professionals elsewhere about the underpinning SSNAP data ⁶⁶ it seems important to ensure that intended adopters are reassured about the integrity of modelling based on this data. Second, evidence from this research and elsewhere indicates that the ED physicians' may have less confidence in the evidence base for, and safety of thrombolysis ^{75, 79}. It is therefore likely that more work will need to be done with the ED physician community to build trust in the SAMueL-2 technology: recruiting ED physicians as brokers/clinical champions may address this. Third, perceived lack of funding/resource and stroke workforce shortages ^{89, 90} may impede quality improvement and adoption of new technologies such as SAMueL-2. It will therefore be vital to address these concerns to ensure sustained use and adoption.

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